



PSTAT 5A: Homework 01

Summer Session B 2025, with Annie Adams

1. Consider the list of numbers $X = \{-3, -1, 0, 0.4, 0.7, 3.9, 6\}$.

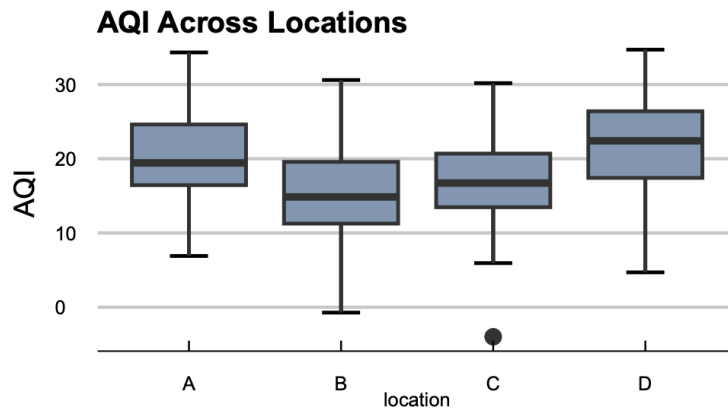
(a) Compute the mean of X .

(b) Compute $\text{median}(X)$, the median of X .

(c) Compute the standard deviation of X .

(d) Compute the range of X .

2. A researcher has collected measurements on the AQI (Air Quality Index) at several locations in Santa Barbara. She has labeled her locations A through D, and collected 100 AQI measurements from each location. The results of her study are displayed graphically below:



- (a) What is the (approximate) median AQI at location A?
- (b) Approximately what percent of AQI readings at Location D were less than 16?
- (c) Are there any outliers at any of the locations?
- (d) From the plot, does it appear that the different locations have different AQI readings? Explain briefly.

3. A recent survey revealed that 40% of UCSB students like Bagel Cafe, 70% like Yetz's Bagels, and 30% like both Bagel Cafe and Yetz's Bagels. A UCSB student is selected at random.

(a) What is the probability that this student likes Bagel Cafe or Yetz's Bagels (*or both*)?

(b) What is the probability that this student likes Bagel Cafe but not Yetz's Bagels?

(c) What is the probability that this student likes neither Bagel Cafe nor Yetz's Bagels?

4. Two numbers are to be selected at random and with replacement from the set $\{1, 2, 3, 4, 5\}$.

(a) Express the outcome space for this experiment using a table.

(b) Express the outcome space for this experiment using a tree.

(c) How many elements are in the outcome space?

(d) For each of the following events, write down the mathematical formulation of the event described (i.e. list out the outcomes comprised in each event):

i. A = the first number selected was 3

ii. B = the second number selected was even

iii. C = the first number selected was 3 and the second number selected was even.

iv. D = the first number selected was odd, and the first number selected was even.

v. E = the first number was strictly greater than the second.

(e) Are we justified in using the Classical Approach to Probability in this problem? Why or why not?

(f) Use the Classical Approach to Probability to compute the probabilities of the events listed out in part (d) above.